LITERATURE REVIEW

Hawkes processes in finance

Emmanuel Bacry, Iacopo Mastromatteo and Jean-François Muzy

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ANALYTICAL ABSTRACT

This survey asks how far the Hawkes process, a point process whose conditional intensity is a linear function of past events, can serve as a common language for high-frequency finance. After defining the process, its stability condition and its first and second order, martingale, scaling and branching properties, the authors organise about thirty applications by theme: trade and price clustering, the measurement of market endogeneity through the branching ratio, tick-level price models, market impact and optimal execution, level-I and full order book models, jump contagion, credit risk, news and high-dimensional networks. Two stylised facts recur across the studies: fitted kernels decay as power laws with an exponent near zero, and the branching ratio sits close to its critical value of one. This review reproduces the key formulas, derives the variance-based branching-ratio estimator, and simulates it to show that it converges only when the counting window greatly exceeds the correlation time. The main limitation is identification: self-excitation cannot be separated from exogenous non-stationarity without further structure, so reading the branching ratio as market reflexivity remains a modelling statement rather than an established causal fact.

1 Introduction

1.1 Paper identification and classification

The paper is a survey of the academic literature on Hawkes processes in high-frequency finance, written by two applied mathematicians and a physicist at the Centre de Mathématiques Appliquées of École Polytechnique and at the University of Corsica. The version read in full is arXiv 1502.04592v2, 48 pages including the appendices and the bibliography of 70 items. The field is quantitative finance and the subfield is market microstructure and financial econometrics of point processes. The paper type is a review, but with a substantial methodological core: Section 2 and Appendices B and C are a self-contained mathematical treatment of the process, its simulation and its estimation, so the paper functions both as a literature map and as a technical reference.

1.2 Executive understanding

WHAT IS THE PAPER TRYING TO DISCOVER?

PAPER The authors do not test a hypothesis. They ask which problems of high-frequency finance have been addressed with Hawkes processes, what the models found, and which methodological tools make them usable. Implicitly they argue that a single object, the linear self-exciting point process, unifies questions that had been treated by separate models.

WHY DOES THE QUESTION MATTER?

Transaction-level data are discrete events, not continuous prices. Duration models of the autoregressive conditional duration type (Engle and Russell, 1998) describe the gaps between events but are awkward in multivariate settings and do not give a transparent notion of one event type causing another. A conditional-intensity description handles several event types jointly, and a linear intensity gives closed-form moments, a branching interpretation, and tractable estimation. If the claim of the survey is right, a market's endogeneity, its stability, its price impact and its order book resilience can all be read from kernel norms and shapes.

WHAT WAS KNOWN BEFORE, AND WHAT IS MISSING?

Before the survey, three separate threads existed: seismology, where Hawkes (1971) and Ogata (1988) introduced the process; econometric point-process models of trades (Hasbrouck, 1991; Engle and Russell, 1998; Bowsher, 2007); and the microstructure literature on long-memory order flow and price impact (Bouchaud, Farmer and Lillo, 2008). What was missing was a consolidated account that connects the mathematics (stability, scaling limits, cumulants, Wiener-Hopf identification) to the empirical findings and exposes the common stylised facts.

WHAT DOES THE PAPER PROPOSE, HOW DOES IT INVESTIGATE, AND WHAT DOES IT FIND?

PAPER The paper proposes a thematic taxonomy. It reviews the theory in Section 2, then applications by theme in Sections 3 to 7, and tabulates the empirical studies in Appendix A. It finds, by aggregation across studies and not by new estimation, that fitted kernels are slowly decaying, that the branching ratio of price and trade events is close to one, that market order flow is mostly self-excited, that price changes are mostly cross-excited and mean-reverting, and that a Hawkes approach can reproduce signature plots, the Epps effect, impact curves and spread distributions. CRITIQUE The paper contains no original estimation. Every number is quoted from a primary study, so the review must be read as an organised secondary source, and its conclusions are only as strong as the studies it selects.

2 Research Problem and Hypothesis

2.1 Research question

PAPER The paper asks which empirical regularities of high-frequency markets can be captured by a multivariate counting process whose intensity is linear in past events, and which analytical and statistical properties of that process make such capture reliable. In the review's own terms: does the Hawkes framework, with kernels estimated from data, deliver a coherent and interpretable description of event clustering, endogeneity, impact and order book dynamics across assets, venues and periods?

2.2 Hypotheses

The survey states no formal hypotheses; the following are implicit in its organisation.

- Event arrivals in electronic markets are self- and cross-exciting, and the interaction is summarised by a non-negative kernel matrix with spectral radius below one.
- The relevant kernels have power-law tails with exponent close to zero, which implies near-critical behaviour and long memory in order flow.
- The branching ratio is a measure of the share of events that are endogenous, and so of market reflexivity.
- Impact, order book resilience and contagion can be represented by cross-excitation blocks of the same kernel matrix.

2.3 Existing literature and research gap

PAPER The authors position Hawkes models as a continuous-time alternative to aggregated discrete-time models and to ACD models. They note that Bowsher (2007) first used a bivariate Hawkes process on trades and mid-price changes, and that the decade after 2005 produced a diversity of applications with little cross-reading.

CRITIQUE The gap is real but modest and mostly organisational. The survey does not discuss competing approaches in depth: Cox processes with stochastic exogenous intensity, duration models with stochastic volatility, or zero-intelligence and agent-based order book models appear only by name or in passing. A reader therefore cannot judge from the paper how much of the empirical success is specific to Hawkes dynamics and how much follows from any model with persistent intensity.

2.4 Claimed contribution

CRITIQUE The contribution is threefold: a unified mathematical summary, a thematic map of applications, and a methodological guide to simulation and estimation. The first and the third are demonstrated well, since the theorems are stated with attribution and the estimation appendix gives the competing methods and their domains of validity. The second is demonstrated only in the sense of coverage; the paper does not provide selection criteria, does not discuss null results or failed fits, and so cannot support strong claims about the generality of the stylised facts.

3 Theoretical Framework and Mechanism

3.1 Definition and intensity

PAPER Let $N_t = (N_t^1, \dots, N_t^D)$ be a vector of counting processes with filtration $\mathcal{F}_t$ containing the history strictly before t. The conditional intensity is

$$\lambda_t = \lim_{\Delta \to 0} \Delta^{-1} \, \mathbb{E}[N_{t+\Delta} - N_t \mid \mathcal{F}_t] \, . \tag{1}$$

N_t^i Cumulative number of events of type i up to time t, with $N_0 = 0$.

λ_t^i Instantaneous probability per unit time of an event of type i, given the past.

A Hawkes process specifies this intensity as a linear functional of the past,

$$\lambda_t^i = \mu^i + \sum_{j=1}^{D} \int_{-\infty}^{t} \varphi_{ij}(t - t') \, \mathrm{d}N_{t'}^j = \mu^i + \sum_{m \colon t_m < t} \varphi_{i,k_m}(t - t_m), \tag{2}$$

or in compact form $\lambda_t = \mu + \Phi * \mathrm{d}N_t$.

μ^i Exogenous (baseline) intensity of type i, the immigrant rate.

$\varphi_{ij}(t)$ Kernel: extra intensity of type i at lag t caused by one event of type j. It is non-negative, causal ($\varphi_{ij}(t) = 0$ for $t < 0$) and integrable.

(t_m, k_m) Time and type of the m-th event.

The equation says that each past event adds a bump to the future intensity, and bumps add. Non-negativity ensures $\lambda \geq 0$ and also forbids inhibition, which is the price of linearity. The authors state that the process has asymptotically stationary increments if the spectral radius of the matrix of kernel norms $\|\Phi\| = \{\|\varphi_{ij}\|_1\}$ is below one (condition H).

3.2 Branching interpretation and stability

PAPER Because the intensity is linear, the process has an equivalent cluster representation (Proposition 6). Immigrants of type i arrive as a Poisson process of rate μ^i. Each event of type j at time s independently produces children of type i as an inhomogeneous Poisson process of rate $\varphi_{ij}(t-s)$, and the process iterates over generations. The expected number of children of type i from one event of type j is $\int_0^\infty \varphi_{ij} = \widehat{\varphi}_{ij}(0)$. In dimension one, writing $n = \|\Phi\| = \int_0^\infty \varphi$, the genealogy is a Galton-Watson tree with mean offspring n. Three phases follow: sub-critical for $n < 1$ (finite clusters), critical for $n = 1$ (finite but heavy-tailed cluster sizes, divergent mean number of generations), and super-critical for $n > 1$ (positive probability of infinite progeny).

INFERENCE A derivation of the mean intensity shows why n is also a share. Taking expectations in Equation (2) in the stationary state with $D = 1$ gives $\Lambda = \mu + n\Lambda$, so

$$\Lambda = \frac{\mu}{1-n}, \qquad \frac{\mu}{\Lambda} = 1 - n, \qquad \frac{\Lambda - \mu}{\Lambda} = n. \tag{3}$$

The fraction of events that are not immigrants is n, which is the sense in which Filimonov and Sornette (2012) read n as a reflexivity index. Each immigrant starts a cluster with expected size $1/(1-n)$ events including itself, hence $n/(1-n)$ descendants. At $n = 0.9$ a cluster contains ten events on average and at $n = 0.95$ twenty, which is the quantitative content of the statement that markets are “close to critical”.

3.3 Exponential kernels and the Markov property

PAPER For $\varphi_{ij}(t) = \alpha_{ij}\beta e^{-\beta t}\mathbf{1}_{t\geq 0}$ the pair (N_t, λ_t) is Markov. In dimension one with baseline μ the intensity obeys

$$\mathrm{d}\lambda_t = -\beta(\lambda_t - \mu)\,\mathrm{d}t + \alpha\beta\,\mathrm{d}N_t. \tag{4}$$

α Kernel norm, equal to the branching ratio n in dimension one.

β Decay rate; $1/\beta$ is the memory of one bump.

The authors write the equation without the baseline mean-reversion level; the version above is the same equation expressed around μ. Between events the intensity relaxes exponentially to μ, and each

event kicks it up by $\alpha\beta$. The Markov property is why the likelihood costs $O(MD)$ for M events instead of $O(M^2D)$ (Appendix C.1) and why exact simulation is cheap.

For the power-law kernel the paper uses

$$\varphi(t) = \frac{\alpha\beta}{(1+\beta t)^{1+\gamma}}, \qquad \|\varphi\| = \frac{\alpha}{\gamma}, \qquad \Lambda = \mu\,\frac{\gamma}{\gamma-\alpha} \quad (\alpha<\gamma). \tag{5}$$

γ Tail exponent. Small γ means very long memory.

α/γ Branching ratio; stability requires $\alpha<\gamma$.

With $\gamma \to 0$ the stability window $\alpha < \gamma$ becomes extremely narrow, which already hints that the empirical finding $\gamma \simeq 0$ is delicate.

3.4 Second-order structure and kernel inversion

PAPER Define the resolvent kernel Υ as the causal solution of $\Phi + \Upsilon * \Phi = \Upsilon$, that is

$$\Upsilon = \Phi + \Phi*\Phi + \Phi*\Phi*\Phi + \cdots, \qquad \hat{\Upsilon}(z) = (I - \hat{\Phi}(z))^{-1} - I, \tag{6}$$

with Laplace transform $\hat{f}(z) = \int f(t)e^{zt}\mathrm{d}t$. The series is the sum over generations of the cluster: $\Upsilon_{ij}(t)$ is the expected rate of type-i descendants at lag t of a type-j event. Then (Proposition 4)

$$\Lambda = (I + \hat{\Upsilon}(0))\mu, \qquad \hat{c}(z) = (I + \hat{\Upsilon}(-z))\,\Sigma\,(I + \hat{\Upsilon}^T(z)), \quad \Sigma_{ii} = \Lambda_i, \tag{7}$$

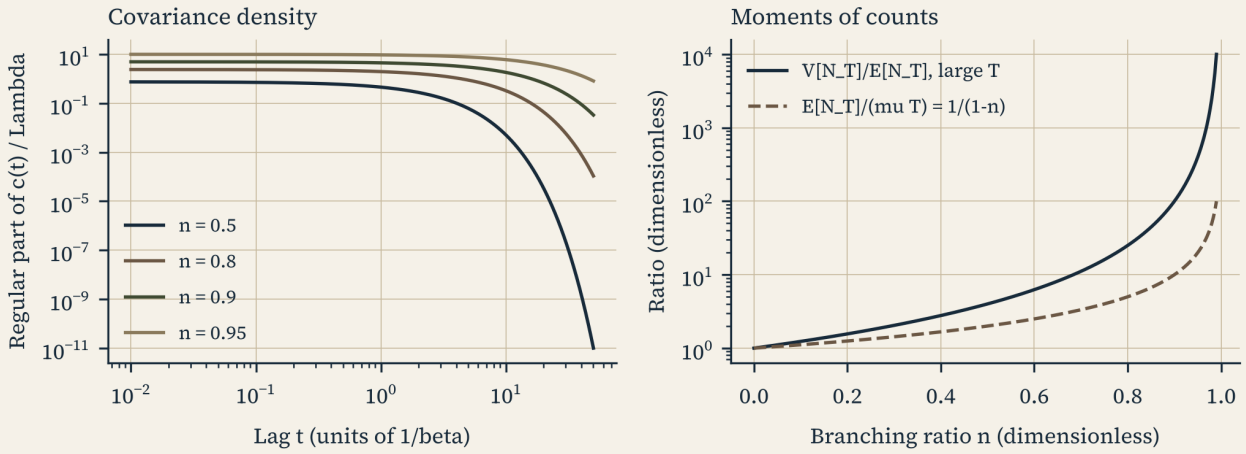
where $c(t-t') = \mathbb{E}[\mathrm{d}N_t\,\mathrm{d}N_{t'}^T]/\mathrm{d}t\,\mathrm{d}t' - \Lambda\Lambda^T$ is the lagged covariance density. Equation (7) gives mean and covariance as functions of (μ, Φ). It also yields the diagonal Σ because the point process variance equals its mean at zero lag, $(\mathrm{d}N)^2 = \mathrm{d}N$. In dimension one it reduces to $\hat{c}(z) = \Lambda/[(1-\hat{\varphi}(-z))(1-\hat{\varphi}(z))]$.

INFERENCE For the exponential kernel $\varphi = \alpha\beta e^{-\beta t}$, $\hat{\varphi}(z) = \alpha/(1-z/\beta)$ and the covariance density is

$$c(t) = \Lambda\left[\delta(t) + \frac{\alpha\beta(2-\alpha)}{2(1-\alpha)}\,e^{-(1-\alpha)\beta|t|}\right]. \tag{8}$$

This is the one-dimensional case of Equation (29) in the paper, in which the bivariate symmetric mode has $\alpha^{(s)} \pm \alpha^{(c)}$ in place of α. The regular part integrates to $\alpha(2-\alpha)/(1-\alpha)^2 = (1-\alpha)^{-2} - 1$, which is consistent with $\hat{c}(0) = \Lambda/(1-\alpha)^2$. The decay rate of correlations is $(1-\alpha)\beta$, so the correlation time diverges as $\alpha \to 1$, even though each individual bump decays at the fixed rate β. Figure 1 shows this. For a power-law kernel the paper (Equation 34) reports $c(t) \sim (\beta t)^{-\gamma-1}$ for $\gamma < 1$ and states an intermediate scaling $c(t) \sim t^{2\gamma-1}$ near criticality for $\gamma < 1/2$, which corresponds to an apparent Hurst exponent $H = 1/2 + \gamma$.

Figure 1. **Exponential Hawkes process: covariance density and moments of counts**



Source: generated by the reviewer from Equation (8) and from the large-window moments $\mathbb{E}[N_T] = \Lambda T$, $\mathbb{V}[N_T] \simeq \Lambda T/(1-n)^2$ with $\Lambda = \mu/(1-n)$. Left: regular part of $c(t)/\Lambda$ with $\beta = 1$. Right: variance and mean of counts, normalised by μT and by $\mathbb{E}[N_T]$ respectively. No empirical data.

3.5 Identification of the kernel: the Wiener-Hopf system

PAPER Define the conditional intensity matrix for $t > 0$ by $g_{ij}(t) = \mathbb{E}[\mathrm{d}N_t^i \mid \mathrm{d}N_0^j = 1]/\mathrm{d}t - \Lambda_i$. From Equation (7), $c(t) = \Sigma g^T(t)$. Theorem 2 states that Φ is the unique causal integrable solution χ of

$$g(t) = \chi(t) + \int_0^t \chi(t-s)\, g(s)\, \mathrm{d}s, \qquad t > 0. \tag{9}$$

$g(t)$ Excess intensity at lag t after one event, estimated from data.

$\chi(t)$ Unknown kernel matrix; the theorem identifies it with Φ.

The equation says that the observed total response g is the direct response χ plus the response to the responses; it is Equation (6) rewritten for observables. The consequence is that means and second-order moments determine the kernel uniquely, which is what licenses non-parametric estimation. The authors add that existence can fail when inhibition is important, because a linear Hawkes process with non-negative kernel cannot reproduce every admissible covariance.

3.6 Martingale representation, prediction and scaling limits

PAPER The compensated process $Y_t = N_t - \int_0^t \lambda_s \, \mathrm{d}s$ is a martingale (Theorem 3) and the intensity satisfies

$$\lambda_t = \mu + \int_0^t \Psi(t-s)\mu \, \mathrm{d}s + \int_0^t \Psi(t-s)\, \mathrm{d}Y_s, \tag{10}$$

$$\mathbb{E}[\lambda_t \mid \mathcal{F}_s] = \mu + \int_0^t \Psi(t-u)\mu \, \mathrm{d}u + \int_0^s \Psi(t-u)\, \mathrm{d}Y_u, \quad t > s. \tag{11}$$

The second expression is the forecast: the intensity at a future date is the stationary mean plus the decaying echo of past surprises $\mathrm{d}Y$, filtered by Υ. This is the basis of the prediction used by Hewlett (2006) for execution.

For large horizons the authors cite Bacry, Delattre, Hoffmann and Muzy (2013b). Under condition H, $\sup_{u\in[0,1]}\|T^{-1}N_{uT}-u\Lambda\|\to 0$ almost surely, and if $\int t^{1/2}\varphi_{ij}<\infty$,

$$T^{1/2}\left(T^{-1}N_{uT}-u\Lambda\right)\xrightarrow[T\to\infty]{\text{law}}\left(I+\|\Psi\|\right)\Sigma^{1/2}W_u. \tag{12}$$

For the bivariate exponential example the limit for the combination $N^{\pm}$ has variance $\Lambda_0 T/(1-\alpha^{(s)}\mp\alpha^{(c)})^2$, consistent with the factor $(1-n)^{-2}$ obtained above. In dimension one the asymptotic variance of counts is therefore inflated by $(1-n)^{-2}$ relative to the Poisson value ΛT.

Near criticality, Jaisson and Rosenbaum (2015) take a sequence of kernels $\varphi^{(T)}=a_T\varphi$ with $\|\varphi\|=1$ and $T(1-a_T)\to\kappa>0$ and show that

$$Z_u^{(T)}=\frac{1-a_T}{T}N_{uT}^{(T)}\implies\text{an integrated Cox-Ingersoll-Ross process}, \tag{13}$$

so a nearly unstable short-memory Hawkes process becomes a square-root volatility process at large scales. The authors note that this framework requires a kernel tail exponent above one, which is incompatible with the empirical $\gamma\simeq 0$ for trades, and that Jaisson (2014) treats the power-law case with $\gamma\in]0,1/2[$, obtaining a power-law decay of correlations with exponent $1-2\gamma$. Proposition 3 and Theorem 1 complete the picture: a critical short-range univariate Hawkes process is degenerate, while a non-trivial critical process exists only for $\gamma\in]0,1/2[$.

3.7 Non-linear and marked extensions

PAPER Marks ξ enter as $\lambda_t^i=\mu^i+\sum_m\varphi_{i,k_m}(t-t_m,\xi_m)$, typically with $\varphi_{ij}(t,\xi)=\varphi_{ij}(t)\chi_{ij}(\xi)$; non-linearity enters as $\lambda_t^i=h(\mu^i+\sum_j\int\varphi_{ij}\,\mathrm{d}N^j)$. The first is used for trade volumes and drawdown sizes, the second to allow inhibition through negative kernels, at the cost of losing most closed-form results. Time-varying baselines $\mu(t)$ handle intraday seasonality.

3.8 Applications as model blocks

PAPER Four constructions from the applications sections are central. In the price model of Bacry, Delattre, Hoffmann and Muzy (2013a) the mid-price is $P_t=P_0+N_t^1-N_t^2$ with (N^1,N^2) the up and down tick processes and a kernel matrix with self part $\varphi^{(s)}$ and cross part $\varphi^{(c)}$; the mean-reverting scenario sets $\varphi^{(s)}=0$. In the Hawkes impact model (HIM) of Bacry, Iuga, Lasnier and Lehalle (2014), with a buying meta-order at trading rate $r(t)$,

$$\lambda_t^1=\mu+\varphi^{(s)}*\mathrm{d}N_t^2+\varphi^{(I)}*f(r_t),\qquad\lambda_t^2=\mu+\varphi^{(s)}*\mathrm{d}N_t^1+\varphi^{(x)}*f(r_t), \tag{14}$$

where f is the instantaneous impact function, $\varphi^{(I)}$ the impact kernel and $\varphi^{(x)}$ the cross-impact kernel. The impulsive version sets $\varphi^{(I)}=\delta$ and $\varphi^{(x)}=C\varphi^{(s)}/\|\varphi^{(s)}\|$. The constant C compares the contrarian

reaction (price reverting) with the herding reaction: $C = 0$ gives a permanent impact, $C = 1$ gives full decay, and $C \in]0, 1[$ gives partial decay; for a power-law $\varphi^{(s)} \sim t^{-\gamma}$ the impact decays to its permanent value as $t^{-(\gamma+1)}$.

In the mutually exciting jump-diffusion of Aït-Sahalia, Cacho-Diaz and Laeven (2015) the log-price vector follows

$$\mathrm{d}X_t = \mu_t \,\mathrm{d}t + \sigma_t \,\mathrm{d}W_t + Z_t \,\mathrm{d}N_t, \tag{15}$$

with N_t a D-variate Hawkes process so that jumps in one market raise the jump intensity in others. Rambaldi, Pennesi and Lillo (2015) model news with a deterministic exogenous kernel,

$$\lambda_t = \mu + \varphi * \mathrm{d}N_t + \varphi^{(n)}(t - t_0), \tag{16}$$

where $\varphi^{(n)}$ may be non-zero for $t < t_0$ to capture anticipation, and its norm measures the news effect. For high-dimensional networks Linderman and Adams (2014) use

$$\varphi_{ij}(t) = A_{ij} W_{ij} h_{\theta_{ij}}(t), \qquad \int h_\theta = 1, \tag{17}$$

with A a random binary adjacency matrix drawn from a latent-distance graph prior, W non-negative weights and h_θ a logistic-normal density.

4 Data and Empirical Strategy

4.1 What counts as data in a survey

PAPER The paper has no dataset of its own. Appendix A tabulates 27 studies, listing the dimension D, the event type N_t, the contracts, the sample length T, the time resolution $\mathrm{d}t$ and the kernel shape. Table 1 reproduces a subset that illustrates the range. The numbers are copied from the paper's Appendix A and have not been verified against the primary studies.

Table 1. Selected empirical studies in Appendix A of the survey

Study	D	Events	Contract	Sample, resolution	Kernel
Bacry, Dayri, Muzy (2012)	1	Trades	Futures	3.5 months, 1 ms	Non-parametric
Bacry and Muzy (2014a)	1	Trades	Futures	4.5 years, 1 ms	Non-parametric
Filimonov, Sornette (2012)	1	Mid-price	Futures	13 years, 1 s	Exponential
Hardiman et al. (2013)	1	Mid-price	Futures	14 years, 1 ms	Sum of exponentials
Bacry et al. (2013a)	2	Price	Futures	2 hours, 1 ms	Exponential
Bacry and Muzy (2014b)	4	Price and trades	Futures	3.75 years, 1 ms	Non-parametric
Large (2007)	10	Level-I book	Stocks	22 days, 1 s	Exponential
Bormetti et al. (2015)	1×20	Price co-jumps	Stocks	88 days, 1 min	Exponential
Linderman, Adams (2014)	100	Extreme returns	Stocks	1 week, 1 s	Exponential, fixed decay
Errais et al. (2010)	1	Defaults	Credit	21 days, 1 day	Exponential

Source: copied from Appendix A of Bacry, Mastromatteo and Muzy (2015) (official). The reviewer selected the rows; the abbreviations for kernel shape follow the paper. Sample lengths are as printed there.

CRITIQUE Three features of this collection matter for inference. First, samples vary from two hours to 17 years and from 1 microsecond to one day, so no single stylised fact can be supported by more than a handful of comparable studies. Second, the data are dominated by liquid index and bond futures (Euro-Bund, Euro-Stoxx, E-mini S&P500) and a few European stocks, and the few FX studies use throttled platform data at 0.1 s resolution, which requires pre-processing (Lallouache and Challet, 2014). Third, most data are 2000 to 2012 vintage, before the major regulatory and technological changes that followed.

4.2 Estimation methods the studies rely on

PAPER Appendix C separates parametric and non-parametric estimation. Maximum likelihood (Ogata, 1981) maximises

$$\log L(\mu, \Phi) = -\sum_{i=1}^{D} \int_0^T \lambda_t^i \, \mathrm{d}t + \sum_{m=1}^{M} \log \lambda_{t_m}^{k_m}, \qquad (\mu^*, \theta^*) = \arg\max_{\mu,\theta} \log L(\mu, \Phi_\theta). \tag{18}$$

$\int \lambda$ Compensator: expected number of events; its negative is the likelihood cost of the empty time between events.

$\log \lambda_{t_m}$ Reward for assigning high intensity to the times where events occur.

The cost is $O(M^2 D)$ for a general kernel and $O(MD)$ for exponentials, which is why exponentials dominate in practice. Expectation-maximisation uses the cluster representation, alternating between the posterior probability $p_{mm'}$ that event m is the parent of event m' and parameter updates; it converges quickly only when $\Lambda\tau \ll 1$, with τ the kernel support. For high dimension, the authors report L^1 and trace-norm penalised likelihoods with fixed β.

The L^2 contrast approach (Reynaud-Bouret and Schbath, 2010; Hansen, Reynaud-Bouret and Rivoirard, 2015) minimises

$$C(\mu, \Phi) = \sum_{i=1}^{D} \left(\int_0^T \lambda_t^i(\mu, \Phi)^2 \, \mathrm{d}t - 2 \int_0^T \lambda_t^i(\mu, \Phi) \, \mathrm{d}\tilde{N}_t^i \right), \tag{19}$$

whose expectation satisfies

$$\arg\min \mathbb{E}[C] = \arg\min \sum_i \mathbb{E}\left[\left(\lambda^i(\mu, \Phi) - \tilde{\lambda}^i \right)^2 \right], \tag{20}$$

where $\tilde{N}$ and $\tilde{\lambda}$ are the true process and its true intensity. The reason is that $\mathbb{E}[\mathrm{d}\tilde{N}_t^i \mid \mathcal{F}_t] = \tilde{\lambda}_t^i \mathrm{d}t$, so the cross term is the covariance with the truth. The Wiener-Hopf method (Bacry and Muzy, 2014a) estimates Λ from counts, estimates $g(t)$ by kernel density, discretises Equation (9) on quadrature points and inverts it, then recovers μ from Equation (7). Simulation uses Ogata's (1981) thinning, time change, or the cluster algorithm of Equation (6).

4.3 Identification: where does the variation come from?

CRITIQUE The identifying variation is the time series of event timestamps itself. Two features need to be distinguished. In the parametric case, identification of (μ, Φ) rests on the correct specification of the intensity, because a mis-specified kernel is absorbed into μ or into the shape of φ. In the non-parametric Wiener-Hopf case, identification is by second-order moments and is exact under the assumption that the data are generated by a stationary linear Hawkes process, but a non-stationary baseline $\mu(t)$ also creates covariance in the counts and cannot be told apart from self-excitation unless $\mu(t)$ is separately modelled. This is the central identification problem for the interpretation of n as endogeneity, and it is developed in Section 7.

4.4 Data limitations and biases

- Selection of studies: the survey does not state search criteria and reports mainly positive fits, so publication bias cannot be excluded. Failed fits are mentioned only in passing (for instance Lallouache and Challet, 2014, on a whole trading day).
- Time resolution and timestamps: throttled data, clock synchronisation and ties between events affect kernel estimates at short lags, which is where the largest mass of the kernel sits.
- Intraday seasonality and overnight gaps: studies truncate to windows such as 9:30 to 11:30 (Muni Toke and Pomponio, 2012) or one hour averaged across days, which shortens the sample and makes branching-ratio estimates sensitive to the baseline.
- Look-ahead and survivorship are not an issue in the usual sense because events are modelled in real time, but the contract choice (front contracts, liquid stocks) is a selection on liquidity.
- Data mining: with several event definitions, kernel families and window choices per study, the number of effective specifications is large and rarely reported.

5 Results

5.1 Near-critical, power-law kernels in trade and price events

PAPER Bacry, Dayri and Muzy (2012) calibrate a one-dimensional non-parametric model to Euro-Bund future trades over 75 trading days of 2009 and find that the kernel is well described over several decades by $\varphi(t) = \alpha\beta(1+\beta t)^{-1-\gamma}$ with $\gamma \simeq 0$ and $\|\Phi\|$ very near one. Hardiman, Bercot and Bouchaud (2013) find a similar shape for E-mini S&P500 mid-price changes, and Bacry and Muzy (2014a) for Euro-Stoxx trades. Figure 6 of the survey overlays the kernels from two of these studies and shows them nearly identical across markets and periods. Da Fonseca and Zaatour (2014) fit exponential kernels to market-order flow and obtain $\beta \in [0.02, 0.1]$ and $\alpha \geq 0.9$, often ≥ 0.95.

EVIDENCE The data establish that the arrivals of trades and mid-price changes are strongly clustered and that fitted exponential kernels need a very small decay rate and a near-unit norm. CRITIQUE They do not by themselves establish a power law, since a short sum of exponentials can mimic one over a finite range. INFERENCE Using Equation (3), $\alpha = 0.95$ implies a cluster of twenty events per immigrant and a correlation time of $1/((1-\alpha)\beta) = 20/\beta$. With $\beta = 0.05$ s^{-1} this is 400 seconds. This arithmetic is the reviewer's own and is illustrative.

5.2 Is market reflexivity increasing?

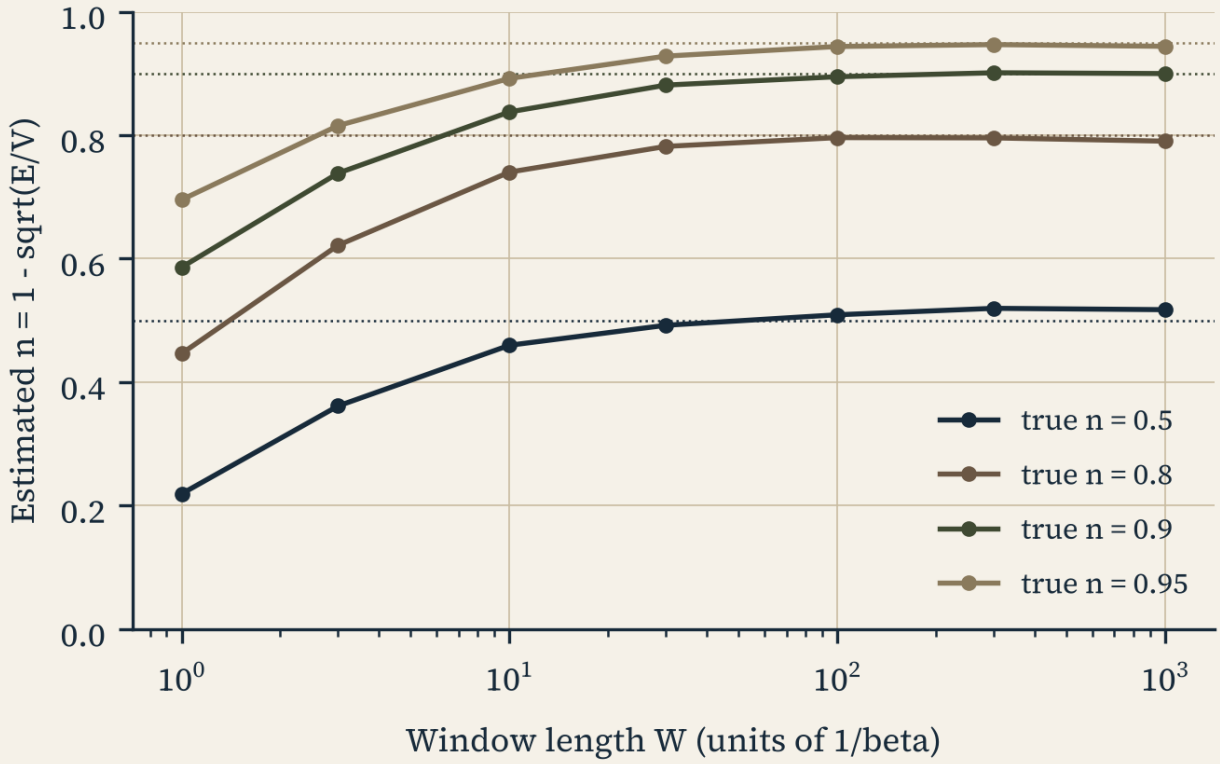
PAPER Filimonov and Sornette (2012) fitted an exponential Hawkes process to E-mini S&P500 mid-price changes over 1998 to 2010 and found an increase in n, attributed to algorithmic trading. Hardiman, Bercot and Bouchaud (2013) showed that the exponential parametrisation is biased when the true kernel is a power law, and that with a power-law kernel n is constant and close to one. Filimonov and Sornette (2013) catalogue the pitfalls (outliers, edge effects, non-stationarity), and Hardiman and Bouchaud (2014) propose the model-free approximation

$$\|\Phi\| \simeq 1 - \sqrt{\frac{\mathbb{E}[N_T]}{\mathbb{V}[N_T]}}, \tag{21}$$

for large T. The printed formula of the paper is typeset ambiguously; the form above is the one implied by Equation (7) and was confirmed by simulation below.

INFERENCE Derivation of Equation (21): in dimension one, $\hat{c}(0) = \Lambda/(1-n)^2$, and for large T, $\mathbb{V}[N_T] \simeq T\hat{c}(0)$ with $\mathbb{E}[N_T] = \Lambda T$, so $\mathbb{V}[N_T]/\mathbb{E}[N_T] = (1-n)^{-2}$. For a Poisson process the ratio is one and $n = 0$.

Figure 2. **Variance-based estimate of the branching ratio as a function of the window length**



Source: generated by the reviewer. Simulation of a univariate exponential Hawkes process by Ogata thinning with $\mu = 1$, $\beta = 1$, horizon $T = 2 \times 10^5$ (units of $1/\beta$), random seed fixed, one realisation per n. Number of events: 0.40, 1.00, 1.99 and 3.98 million for $n = 0.5$, 0.8, 0.9, 0.95. Dotted lines mark the true value of n.

EVIDENCE Figure 2 (the reviewer's simulation, Table 2) shows that the estimator is biased downward for windows comparable to the correlation time and converges from below as W grows, but that the window needed grows as $n \to 1$. For $n = 0.95$ the estimate is 0.70 at $W = 1$, 0.89 at $W = 10$, 0.94 at $W = 100$ and about 0.945 for $W \geq 300$. This matters for the empirical debate: second-resolution counting windows in data with $\beta \sim 0.05$ s^{-1} correspond to $W \simeq 0.05$ in the simulation's units and would produce a strong downward bias, not a near-critical estimate. A near-unit estimate therefore needs windows of many minutes.

Table 2. **Simulation check of the variance-based estimator of the branching ratio**

True n	Mean rate	$W = 1$	3	10	30	100	300	1000
0.50	2.00	0.219	0.361	0.460	0.492	0.509	0.520	0.517
0.80	4.99	0.446	0.622	0.740	0.782	0.796	0.796	0.790
0.90	9.96	0.586	0.739	0.838	0.881	0.895	0.901	0.900
0.95	19.92	0.696	0.816	0.892	0.928	0.944	0.947	0.944

Source: generated by the reviewer; one simulated realisation per row, $\mu = 1$, $\beta = 1$, $T = 2 \times 10^5$; window W in units of $1/\beta$. The mean rate should equal $1/(1 - n)$ (2, 5, 10, 20), which it does to within 0.5%.

5.3 Level-I order book and market impact

PAPER Bacry, Jaisson and Muzy (2015) fit an eight-dimensional model of level-I events (price changes, trades, limit orders and cancellations on each side) to Euro-Bund and DAX futures at microsecond resolution. The matrix of kernel norms is largely diagonal except in the price block, which is anti-diagonal, so that a price move triggers a price move in the opposite direction. Some kernels are negative: for a large-tick asset, an upward move triggers trades on the bid and inhibits trades on the ask. Large (2007) estimates on 22 days of Barclays stock that the book is resilient in fewer than 40% of cases, with replenishment within about 20 seconds when it is.

Bacry and Muzy (2014b) estimate a four-dimensional model with buy and sell market orders and up and down price jumps on Euro-Stoxx and Euro-Bund futures over 800 days. The impact block is nearly diagonal and localised within less than 0.1 seconds, while the feedback block is anti-diagonal with slightly negative diagonal entries. Muni Toke (2010) shows with a two-agent order book model that adding Hawkes self-excitation and a market-to-limit cross kernel improves the fit to inter-arrival times and to the spread distribution, and that the reverse cross kernel is negligible.

Table 3. **Main empirical regularities reported by the survey and strength of the evidence**

Regularity	Reported studies	Magnitude	Reviewer assessment
Trade and price kernels have heavy tails	Bacry et al. (2012); Hardiman et al. (2013); Bacry and Muzy (2014a)	$\gamma \simeq 0$	Consistent across futures; weak outside them
Branching ratio near one	Da Fonseca and Zaatour (2014); Hardiman et al. (2013)	$\alpha \geq 0.9$ to 0.95	Sensitive to estimator and baseline
Order flow self-excited	Bowsher (2007); Large (2007); Muni Toke (2010)	Cross terms small	Plausible; meta-order splitting is a hypothesis
Price changes anti-diagonal, mean-reverting	Bacry et al. (2015); Bacry and Muzy (2014b)	Negative cross-excitation	Consistent; tick-size dependent
Impact is short-range in the kernel, long-range through order flow	Bacry and Muzy (2014b); Bacry et al. (2014)	Impact localised below 0.1 s	Central for impact; permanent impact unresolved
Contagion and clustering of extremes	Aït-Sahalia et al. (2015); Embrechts et al. (2011)	Significant self-excitation	Daily or hourly data, fewer events

Source: reviewer's synthesis of statements in Sections 3 to 7 of Bacry, Mastromatteo and Muzy (2015); the assessment column is the reviewer's.

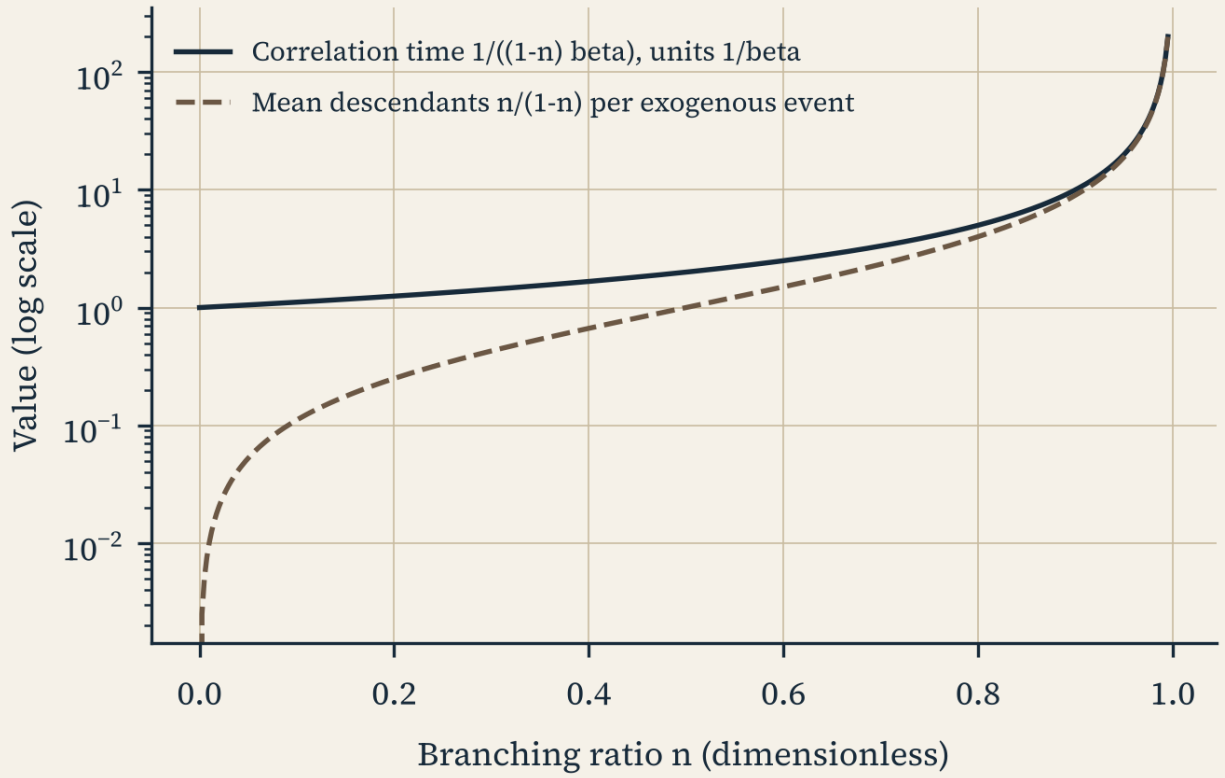
5.4 Figure analysis: Figure 6 of the survey and the reviewer's figures

EVIDENCE Figure 6 of the survey plots the inferred kernel $\varphi(t)$ in log-log scale for E-mini mid-price changes (1998, 2006, 2009) and Euro-Stoxx trade arrivals, from 10^{-3} to 10^{2} seconds, with values from 10^{-5} to 10^{2}. The relationship visible is a nearly straight line over four to five decades, with a cut-off at short lags. It is associational: it shows that the estimated function is similar across datasets, not that the same mechanism generates it. Different estimators were used (parametric sum of exponentials

in one study, Wiener-Hopf in the other), which strengthens the similarity claim but leaves the common-factor explanation, such as the same pattern of order splitting or news flow, untouched. The unusual observation is the shift of the short-lag cut-off across years, which the authors link to the increase in trading frequency.

EVIDENCE The reviewer's Figure 1 adds the analytic counterpart for the exponential case: with $n = 0.95$ the regular part of the covariance remains above 10^{-1} of its zero-lag value out to lags of order $10/\beta$, while for $n = 0.5$ it is negligible after about $5/\beta$. Figure 3 shows the scale separation that follows: the correlation time grows like $1/(1-n)$, so a 1 percentage point change in n near 0.95 changes the correlation time by roughly 17 to 25%.

Figure 3. **Scale separation as the branching ratio approaches one**



Source: generated by the reviewer from $1/((1-n)\beta)$ and $n/(1-n)$ with $\beta = 1$. Deterministic functions of the exponential Hawkes parameters; no empirical data.

6 Robustness and Additional Tests

CRITIQUE A survey has no robustness section of its own. The relevant question is how far the robustness reported in the primary studies resolves the main concerns. Table 4 assesses the principal ones.

Table 4. **Robustness: concern, test reported, result and assessment**

Concern	Test reported	Result	Does it resolve the concern?
Exponential kernel biases n	Hardiman et al. (2013) refit with power-law and non-parametric kernels	n constant near one instead of rising	Partly: bias is shown, but the stable level also depends on the cut-off and on μ
Outliers, edge effects, non-stationarity	Filimonov and Sornette (2013) analysis of calibration pitfalls	Large biases possible	Partly: identifies the problem, not a fix
Parametric dependence	Hardiman and Bouchaud (2014) model-free variance estimator	Same conclusion	Partly: depends on window and on stationarity (Figure 2)
Goodness of fit	Embrechts et al. (2011); Lallouache and Challet (2014) tests	Good fit for extremes; whole-day fit fails due to seasonality of μ	Partly: reveals the baseline problem
Cross-market validity	Similar kernels for E-mini and Euro-Stoxx (Figure 6)	Near identical shape	Partly: both are liquid futures
Data frequency and throttling	Randomisation of throttled EBS data (Rambaldi et al., 2015)	Effect on news kernel noisy	No: no sensitivity analysis reported in the survey
Existence of linear Hawkes representation with inhibition	Negative kernels, non-linear h (Section 2.2.4)	Estimated negative kernels	Partly: identification of non-linear models weaker

Source: reviewer's reading of Bacry, Mastromatteo and Muzy (2015); the last column is the reviewer's assessment.

CRITIQUE The common pattern is that each robustness check addresses estimator bias, not the specification of the data-generating process. None of the cited tests distinguishes genuine self-excitation from a time-varying exogenous intensity or a latent common factor, which is the concern that matters for the interpretation.

7 Critical Evaluation

7.1 What the paper establishes

ESTABLISHED

The mathematical properties of the linear Hawkes process: the stability condition, the mean and covariance in Equation (7), the cluster representation, the Wiener-Hopf identification, the martingale representation and the Brownian scaling limit are proved in the cited sources and stated precisely. The empirical fact that trade and price event arrivals are strongly clustered and well described by a near-unit-norm slowly decaying kernel in the liquid futures studied is supported by several independent estimations.

SUPPORTED BUT CONDITIONAL

That the market is close to criticality (n near one): conditional on the Hawkes specification, on stationarity within the estimation window, on the choice of kernel family and on a correct treatment of the baseline. That order flow is self-excited because of meta-order splitting: supported by the order-splitting literature (Toth, Palit, Lillo and Farmer, 2015), not tested within a Hawkes model.

SUGGESTIVE

That Hawkes models give a causal decomposition of market events (Proposition 7 and Large, 2007, speaks of precipitation, not cause), that the shape of impact decay is controlled by the kernel exponent (Jaisson, 2014; Bacry et al., 2014), and that the high-dimensional network estimates (Linderman and Adams, 2014; Mastromatteo and Marsili, 2011) reveal economic structure.

NOT ESTABLISHED

That endogenous feedback explains the excess volatility puzzle; that an increase of high-frequency trading raises or does not raise reflexivity; that Hawkes models improve execution costs out of sample; or that parameters are stable enough across regimes for trading or regulation. The survey itself presents several of these as open.

7.2 Strengths

STRENGTHS RECOGNISED BY THE AUTHORS

- Simplicity and flexibility: the process handles marks, non-stationarity and non-linearity, and its parameters have a direct interpretation through the cluster picture.
- Statistical tractability: closed-form moments, fast likelihood for exponentials and non-parametric kernel estimation by inverting second-order moments.

STRENGTHS IDENTIFIED INDEPENDENTLY

- A reliable mathematical core. Equations (6) to (9) allow a reader to compute any second-order prediction of a model without simulation, which makes calibration transparent.
- Clear statement of limits of the theory, for instance the degeneracy of critical short-range processes (Proposition 3) and the incompatibility of the nearly unstable scaling limit with $\gamma \simeq 0$.
- The appendix on estimation states when each method is preferred (slowly decaying kernels, negative kernels), which is practically useful.
- Appendix A creates a comparable table of D, T, resolution and kernel, which allows a reader to see the thin evidence behind some themes.

7.3 Weaknesses and critique

SINGLE MOST IMPORTANT WEAKNESS

Problem. The survey interprets the branching ratio as a measure of endogeneity, without addressing the observational equivalence between self-excitation and exogenous non-stationarity. **Why it matters.** Event clustering can arise from common news arrivals, auctions and scheduled releases, latent volatility and strategy-driven order flow. A Cox process with a persistent stochastic baseline has a covariance structure of the same kind as a Hawkes process with a near-critical kernel; the second-order moments in Equation (7) cannot separate the two when $\mu(t)$ is unobserved. **Potential bias.** Upward bias in n and in kernel tails, with the long memory of the exogenous driver absorbed into a power-law kernel. **How it could be tested.** Estimate the model with a flexible $\mu(t)$ and with exogenous covariates (scheduled announcements, cross-asset activity), compare against a Cox-process benchmark by the time-rescaling residuals, and use shocks that are exogenous by design, such as exchange outages or halts, to measure the response directly.

OTHER WEAKNESSES

- Linearity and positivity: a non-negative kernel cannot represent inhibition, yet the survey itself reports negative kernels. Forcing positivity would bias norms upward. The test is to fit non-linear and unconstrained models and compare norms.
- Stationarity: the theory depends on condition H and on a fixed kernel; the empirical samples span regime changes (tick size, latency). Test by rolling estimation and structural-break tests on n.
- Coverage and selection: the survey states no inclusion rule, shows few negative results, and does not benchmark against non-Hawkes models. Test by a systematic search protocol.
- No quantification of estimation uncertainty: standard errors and confidence bands for n and γ are rarely reported, although near-critical estimates are very sensitive. Test by bootstrap of clusters or by simulation, as in Figure 2.
- Economic content: impact, resilience and contagion results are statistical regularities; the survey does not link kernels to utility, inventory or information models, so it cannot say why the kernel has a given shape.

– Out-of-sample value: forecasts, trading cost reductions and risk measures are mostly absent. Test by out-of-sample intensity forecasts using Equation (11).

8 Contradictions and Edge Cases

CRITIQUE Several questions can be put directly to the framework.

CRITIQUE Near criticality. The mean intensity $\Lambda = \mu/(1-n)$ and the variance $\Lambda/(1-n)^2$ both diverge as $n \to 1$. With $n = 0.95$ and an exogenous rate of one event per second, the mean rate is twenty per second and the cluster has twenty events; a drift of n from 0.95 to 0.99 multiplies the mean rate by five and, at fixed μ, the long-window variance of counts by 125 (using $\mathbb{V}[N_T] = \mu T/(1-n)^3$, which is $8\times10^3\,\mu T$ at $n = 0.95$ and $10^6\,\mu T$ at $n = 0.99$). A model that sits this close to the stability boundary is highly sensitive to small estimation errors in n: an error of 0.01 near 0.99 changes the implied exogenous share $1-n$ by a factor of two. Estimates of n close to one are therefore compatible with a wide range of true exogenous shares, which is a statement about the precision of the quantity and not about the market.

CRITIQUE Power law with $\gamma \simeq 0$. Equation (5) requires $\alpha < \gamma$; with $\gamma \to 0$ the admissible α shrinks to zero, while the kernel mass α/γ stays near one only if α and γ both go to zero. Parameters are then poorly identified separately, and the data constrain essentially the ratio and the cut-off $1/\beta$. In addition, Proposition 3 and Theorem 1 imply that exact criticality with short-range kernels is degenerate and that non-trivial critical processes exist only for $\gamma \in]0, 1/2[$; an empirical $\gamma \simeq 0$ lies at the border of that region. Finite-sample estimates of $\gamma \simeq 0$ may reflect truncation of a long finite memory.

CRITIQUE Scaling limit. The framework of Jaisson and Rosenbaum (2015) turns a nearly unstable short-memory Hawkes process into a CIR volatility, but requires a tail exponent above one while the trade-flow kernel has exponent close to zero. The micro-to-macro link, which is the survey's own stated priority for future work, is therefore not available for the best-documented kernel.

CRITIQUE Alternative mechanisms. Order splitting (Toth et al., 2015), common news arrivals, and the time-of-day pattern of liquidity can each generate positive, slowly decaying autocovariance of event counts. The Hawkes kernel then fits a statistical shape, not a behavioural response. Conversely, strategic traders who *reduce* activity after an event produce inhibitory kernels which a non-negative specification cannot represent.

CRITIQUE Cross-section and regime. The similarity of kernels in Figure 6 holds for two very liquid futures contracts. For stocks with a small tick, throttled FX, or crypto, there is no reason for the near-critical, power-law structure to hold; the survey's own results on FX (Lallouache and Challet, 2014) show worse fits on a full day. A limiting case worth checking is $D \to \infty$: Mastromatteo, Bacry and Muzy (2014) show that near-critical behaviour can emerge from the interaction of many components even with short-range kernels, so criticality of the aggregate does not require long memory of any individual kernel.

9 Further Literature

Table 5. Literature connections

Paper	Shared idea	Relationship	Why it matters
Hawkes (1971)	Linear self-exciting point process	Foundation	Defines the process and its spectral analysis
Ogata (1981; 1988)	Thinning simulation, likelihood, residual analysis	Method foundation	Supplies simulation and estimation tools
Engle and Russell (1998)	Point-process view of trades	Alternative approach	ACD duration models, the main competing framework
Bowsher (2007)	Bivariate Hawkes on trades and mid-price	Preceded	First finance application with explicit intensities
Filimonov and Sornette (2012); Hardiman et al. (2013)	Branching ratio as reflexivity	Contradicting results	Increase versus stable near-critical n
Bouchaud, Gefen, Potters and Wyart (2004)	Propagator and long memory of order flow	Related theory	Gives a non-point-process account of power-law impact decay
Cont, Kukanov and Stoikov (2014)	Order flow imbalance and short-horizon price changes	Different method	Linear regression alternative to a kernel description
Cont and Bouchaud (2000)	Herding and aggregate fluctuations	Related mechanism	Endogenous clustering in a percolation model
Jaisson and Rosenbaum (2015; 2016)	Scaling limits of nearly unstable Hawkes	Extended	Links micro clustering to rough volatility
El Euch and Rosenbaum (2019)	Characteristic function of rough Heston	Applied the limit	Pricing consequences of the scaling limit
Hawkes (2018)	Review of finance applications	Updated review	Covers the literature after 2015
Bacry, Jaisson and Muzy (2016)	Estimation with slowly decaying kernels	Improved estimation	Wiener-Hopf adapted to power-law kernels

Source: reviewer's synthesis. Entries from the survey's bibliography and the reviewer's earlier reviews in this library (Bouchaud et al., Cont et al., Cont and Bouchaud, Jaisson and Rosenbaum, El Euch and Rosenbaum). Hawkes (2018) and Bacry, Jaisson and Muzy (2016) are later works not read in full for this review.

10 Research Gaps

10.1 Paper-level gaps

- **Gap:** No test separating self-excitation from exogenous clustering. *Why it matters:* It determines whether n measures endogeneity. *Existing limitation:* Second-order moments are the same under both. *Way forward:* Natural experiments, exogenous covariates for $\mu(t)$ and Cox benchmarks.
- **Gap:** No uncertainty quantification for n and γ. *Why it matters:* Near-critical estimates are highly sensitive. *Existing limitation:* Few studies report confidence bands. *Way forward:* Cluster bootstrap and simulation-based calibration of estimators.

10.2 Literature-level gaps

- **Gap:** Origin of the power law with exponent near zero. *Why it matters:* It controls criticality and the scaling limit. *Existing limitation:* Order splitting is a conjecture. *Way forward:* Broker data with meta-order identifiers to link the distribution of meta-order lengths to the kernel exponent.
- **Gap:** Micro-to-macro transition. *Why it matters:* It would connect tick dynamics to volatility. *Existing limitation:* The proven scaling limits need tail exponents the data do not show. *Way forward:* Scaling results for $\gamma < 1/2$ and multivariate versions.

10.3 Methodological gaps

- **Gap:** Estimation under inhibition and high dimension. *Why it matters:* Negative kernels are observed. *Existing limitation:* Convex penalised likelihood needs fixed decay and non-negative weights. *Way forward:* Non-linear Hawkes with regularisation, neural point-process models.
- **Gap:** Out-of-sample evaluation. *Why it matters:* Usefulness for trading and risk. *Existing limitation:* Mostly in-sample fit and goodness-of-fit tests. *Way forward:* Forecast comparison of intensities and economic evaluation of execution.

11 Research Opportunities

Idea 1. Exogenous versus endogenous clustering with announcement calendars

RESEARCH QUESTION. How much of the near-critical branching ratio in index futures remains once scheduled macro announcements and cross-asset activity are placed in the baseline?

MOTIVATION. The main weakness of the survey is observational equivalence of self-excitation and exogenous clustering.

HYPOTHESIS. The estimated n falls materially, for instance below 0.8, when $\mu(t)$ is allowed to depend on announcements and common factors.

ECONOMIC MECHANISM. Part of the apparent reflexivity is information arrival and common liquidity demand.

LITERATURE CONNECTION. Filimonov and Sornette (2012); Hardiman et al. (2013); Rambaldi et al. (2015).

DATA. Tick data for E-mini S&P500 and Euro-Stoxx futures with exchange timestamps, plus a calendar of macroeconomic releases. Availability from commercial vendors; not verified in this run.

VARIABLES. Trade and mid-price events, announcement dummies and decay shapes, VIX or cross-asset event counts.

METHOD. Hawkes model with covariate-dependent baseline; Wiener-Hopf and likelihood estimation; time-rescaling residuals; comparison with a Cox process.

IDENTIFICATION STRATEGY. Exogenous timing of scheduled releases; trading halts as shocks.

EXPECTED CONTRIBUTION. A bound on the share of endogenous events.

MAIN RISK. Announcement effects may themselves be partly endogenous, and data licensing is costly.

Idea 2. Tick-size regime changes and the branching ratio

RESEARCH QUESTION. Does a change in the tick size shift the cross-excitation structure of the price block and the branching ratio?

MOTIVATION. Bacry et al. (2015) report that the price-event feedback is very sensitive to the tick size.

HYPOTHESIS. A larger relative tick raises anti-diagonal price cross-excitation and lowers n for mid-price changes.

ECONOMIC MECHANISM. Larger ticks increase queue priority value and mean reversion of the mid-price.

LITERATURE CONNECTION. Bacry et al. (2015); Cont, Kukanov and Stoikov (2014).

DATA. Tick data on treated and untreated stocks around tick-size reforms such as the European 2018 regime and the US pilot programme. Dates and availability would need to be checked from official sources.

VARIABLES. Event types by side, tick-to-price ratio, kernel norms.

METHOD. Difference-in-differences on estimated kernel norms with bootstrap standard errors.

IDENTIFICATION STRATEGY. Regulatory change of tick for a subset of stocks, with matched controls.

EXPECTED CONTRIBUTION. Causal evidence on how market design acts on endogeneity.

MAIN RISK. Estimation noise in kernel norms and concurrent changes in market structure.

Idea 3. Out-of-sample intensity and volatility forecasting

RESEARCH QUESTION. Does the predictor in Equation (11) improve intraday volatility and liquidity forecasts over HAR-type benchmarks?

MOTIVATION. The survey presents no out-of-sample evidence.

HYPOTHESIS. Forecasts based on Hawkes intensities improve the mean squared error at horizons from seconds to minutes, with gains vanishing at daily horizons.

ECONOMIC MECHANISM. Event clustering carries information about the state of the order book not present in past realised variance.

LITERATURE CONNECTION. Hewlett (2006); Bowsher (2007); Hardiman and Bouchaud (2014).

DATA. Intraday event data for liquid futures, 2015 onward.

VARIABLES. Realised variance at several horizons, forecast intensity, order flow imbalance.

METHOD. Rolling estimation, Diebold-Mariano and model confidence set tests.

IDENTIFICATION STRATEGY. Predictive, not causal; strict out-of-sample splits.

EXPECTED CONTRIBUTION. Practical value of Hawkes dynamics.

MAIN RISK. Gains may be absorbed by simple intraday seasonality models.

Idea 4. Sparse low-rank cross-excitation in crypto markets

RESEARCH QUESTION. What is the network of lead-lag excitation among the most traded crypto pairs and perpetual futures, and is it near critical?

MOTIVATION. Continuous trading and fragmented venues offer a setting where network structure and contagion matter.

HYPOTHESIS. A low-rank structure dominated by one market mode, with an aggregate near-critical branching ratio, as in Mastromatteo and Marsili (2011).

ECONOMIC MECHANISM. Arbitrage and liquidation cascades transmit events across venues.

LITERATURE CONNECTION. Linderman and Adams (2014); Mastromatteo, Bacry and Muzy (2014); Ait-Sahalia et al. (2015).

DATA. Public trade and order book feeds from several venues; availability and licence to be confirmed.

VARIABLES. Event counts by venue, side and pair; liquidation flags.

METHOD. Penalised Hawkes likelihood with sparsity and trace-norm penalties; spectral analysis of the kernel norm matrix.

IDENTIFICATION STRATEGY. Predictive and structural: exchange outages as shocks.

EXPECTED CONTRIBUTION. A new market with transparent data.

MAIN RISK. Wash trading and timestamps of uneven quality.

Idea 5. Meta-order splitting and the kernel exponent

RESEARCH QUESTION. Is the kernel exponent γ of market-order flow explained by the distribution of meta-order lengths?

MOTIVATION. The survey leaves the origin of the power law open and suggests splitting.

HYPOTHESIS. A tail exponent of meta-order durations maps into γ through a superposition of renewal processes.

ECONOMIC MECHANISM. Investors split large orders; their persistence generates long memory.

LITERATURE CONNECTION. Toth et al. (2015); Bacry et al. (2014); Jaisson (2014).

DATA. Broker or exchange data with account identifiers, in the spirit of the data used for the HIM paper; access restricted.

VARIABLES. Meta-order size and duration, trade signs, estimated kernel norms.

METHOD. Fit the Hawkes model to anonymous flow and to flow with meta-orders removed; compare kernels.

IDENTIFICATION STRATEGY. Removing identified meta-orders as a quasi-experiment on the origin of persistence.

EXPECTED CONTRIBUTION. Mechanism for the stylised fact.

MAIN RISK. Data access; partial identification of meta-orders.

11.1 Ranking

Table 6. **Ranking of research ideas (scores from 1 to 5, assigned by the reviewer)**

Idea	Novelty	Feasibility	Data	Method	Importance	Contribution	Total
1. Announcement baseline	4	4	3	4	5	5	25
2. Tick size	4	3	3	4	4	4	22
3. Forecasting	3	5	4	4	4	4	24
4. Crypto network	4	4	4	3	3	3	21
5. Meta-order splitting	5	2	1	3	5	5	21

Source: reviewer's subjective scoring; not an official measure.

The top three are ideas 1, 3 and 2. Idea 1 addresses the largest weakness and has direct consequences for the interpretation of the whole literature. Idea 3 is the most feasible and is the only one that tests the practical value of the framework. Idea 2 offers cleaner identification through regulatory variation. Idea 5 is the most interesting scientifically but is limited by data access.

12 Learning Framework

12.1 Prerequisite concepts

- Counting processes, compensators and conditional intensity; Poisson and Cox processes.
- Laplace transforms, convolution and the Neumann series for resolvents.
- Galton-Watson branching processes and criticality.
- Martingales and the functional central limit theorem.
- Limit order book microstructure: market, limit and cancel orders, tick size, price impact and meta-orders.
- Maximum likelihood, the EM algorithm and regularised estimation.

12.2 Core concepts learned

- The kernel norm is a branching ratio and $1 - n$ is the exogenous share.
- Correlation time scales as $1/((1 - n)\beta)$ and counts have variance inflated by $(1 - n)^{-2}$.
- Means and covariances identify the kernel uniquely through the Wiener-Hopf system.
- Slow kernels produce long memory and require non-exponential estimation.
- Cross-excitation blocks encode impact, resilience and contagion.

12.3 Important equations

Equations (2), (3), (6), (7), (9), (11), (21) and (18) are the ones worth memorising: the model, the endogenous share, the resolvent, the moments, the identification, the forecast, the model-free estimator and the likelihood.

12.4 Methods learned

Thinning simulation, cluster simulation, parametric likelihood, EM on the genealogy, L^2 contrast with Lasso, Wiener-Hopf non-parametric estimation, spectral factorisation for symmetric kernels, and variance-based branching ratio estimation.

12.5 Questions to test understanding

COMPREHENSION

1. Why does stability require the spectral radius of $\|\Phi\|$ to be below one, and what does it mean in terms of offspring?
2. What fraction of events is exogenous when $n = 0.9$?
3. Why do exponential kernels allow $O(MD)$ likelihood evaluation?

ANALYSIS

1. Derive Equation (21) from Equation (7).
2. Why is a critical short-range univariate Hawkes process degenerate, and how does a power-law tail change this?
3. Why can the EM algorithm not recover negative kernels?

RESEARCH

1. Construct a Cox process whose covariance matches a Hawkes process and discuss what data would tell them apart.
2. How would you test whether the near-critical branching ratio survives a flexible baseline?
3. What replaces the Hawkes kernel when traders react by withdrawing liquidity?

12.6 Papers to read next

Bacry and Muzy (2014a) on the Wiener-Hopf characterisation; Hardiman, Bercot and Bouchaud (2013); Jaisson and Rosenbaum (2015); Bacry, Jaisson and Muzy (2016); Hawkes (2018).

13 Conclusion

13.1 One-sentence thesis

The linear Hawkes process is a mathematically tractable, empirically flexible description of event clustering in high-frequency markets, with a stylised picture of near-critical, slowly decaying kernels whose economic interpretation remains conditional on the model.

13.2 Three most important insights

- The kernel norm is both a stability criterion and a measure of the endogenous share of events, with mean intensity $\mu/(1-n)$.
- Second-order moments determine the kernel (Wiener-Hopf), which supports non-parametric estimation but ties identification to stationarity and to the baseline.
- Slowly decaying kernels with norms near one are the main empirical regularity and they create instabilities in estimation and in the scaling theory.

13.3 Strongest evidence

The convergence of independent estimations with different estimators and markets on a near-unit, heavy-tailed kernel for trades and mid-price changes (Figure 6 of the survey), together with the proved identification of the kernel from second-order moments.

13.4 Biggest weakness

No separation of self-excitation from exogenous clustering, which undermines the interpretation of the branching ratio as endogeneity.

13.5 Most important research gap

The origin of the near-zero power-law exponent and its relation to meta-order splitting.

13.6 Best research opportunity

Estimating Hawkes models with announcement-driven baselines (Idea 1) to bound the endogenous share.

13.7 Three papers to read next

Bacry and Muzy (2014a); Hardiman, Bercot and Bouchaud (2013); Jaisson and Rosenbaum (2015).

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